

Assessment of RAG Systems in Agentic Workflows: A Comparative Analysis

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Abstract

Retrieval-Augmented Generation (RAG) is becoming a cornerstone for autonomous AI agents. However, stan

1. Introduction

Large Language Models (LLMs) hallucinate when knowledge is missing. RAG mitigates this by providing relev

2. Methods

We developed an Indexer Agent using PyMuPDF for parsing and ChromaDB for vector storage. The system d
The evaluation metric was Hit Rate@5 on a synthetic QA dataset generated from the papers.

3. Results

The Section-Aware method achieved a Hit Rate@5 of 0.85, whereas the Fixed-Window method achieved 0.65.
Qualitatively, the retrieval chunks contained complete arguments rather than fragmented sentences.

4. Discussion

Our findings suggest that structure-preserving parsing is essential for scientific RAG.
Limitations include the reliance on clear PDF headers, which older papers may lack.
Future work will explore vision-based layout analysis.

5. Conclusion

Section-aware chunking significantly boosts RAG performance for scientific literature.